

O(Alpha)

Evaluation Summary and User Testing Readout

Solution suitability, simulated proxy user feedback, and next research plan

Prepared for: O(Alpha) product and submission planning

Date: 8 July 2026

Evidence status: Expert/self evaluation, heuristic walkthrough, and simulated proxy persona review completed

Integrity note: No public launch, real user survey, or actual participant interview has been conducted yet

Recommended claim for the submission

The team used expert/self evaluation, a cognitive walkthrough / heuristic evaluation, and a simulated 10-person proxy persona review. Real participant interviews, low-fidelity tests, surveys, and high-fidelity usability tests are prepared as privacy-safe protocols but should not be marked as completed until fielded with real users.

1. Executive Summary

O(Alpha)'s core paper-trading workflow is understandable and safety-aware: users select a risk profile, run and accept a backtest, launch or terminate a paper agent, and are prevented from changing settings while a portfolio agent is active. The largest product gap is trust calibration. Users need stronger paper-only messaging, visible strategy evidence, and operational freshness indicators before they can confidently judge the system.

- Core workflow health: usable, with average simulated ease of 3.4/5.
- Trust confidence: moderate at 2.9/5, mainly because evidence is not visible enough in the UI.
- Transparency: weakest area at 2.5/5; validation proof is stronger in repo artifacts than in the app screens.
- Safety: active-agent settings lock was the most consistently positive signal, raised by 10/10 proxy personas.
- Top improvement: repeat paper-only / no brokerage orders messaging near Launch Agent, P&L, backtest acceptance, and settings.

2. Evaluation Methods

Method	Status	Select now?	Evidence or next step
Expert / self evaluation	Completed	Yes	Review of docs, frontend flows, safety behavior, and validation story.
Cognitive walkthrough / heuristic evaluation	Completed	Yes	Task walkthrough across onboarding, dashboard, launch/terminate, settings, and evidence discovery.
Simulated user focus group	Completed, simulated	Yes, labelled simulated	Ten proxy personas; six from parallel subagent reviews and four local synthetic personas.
Actual user focus group / interview	Not conducted	No	Private 10-person protocol prepared for future fielding.
Low-fidelity usability testing	Not conducted	No	Figma/wireframe task protocol prepared.
Survey of potential users	Not fielded	No	50-person survey plan and item bank prepared.
High-fidelity usability testing	Not conducted with real users	No	Seeded-account working prototype protocol prepared.

Table 1. Evaluation method status. The actual-user methods are ready to field but not yet completed.

3. Data Snapshot

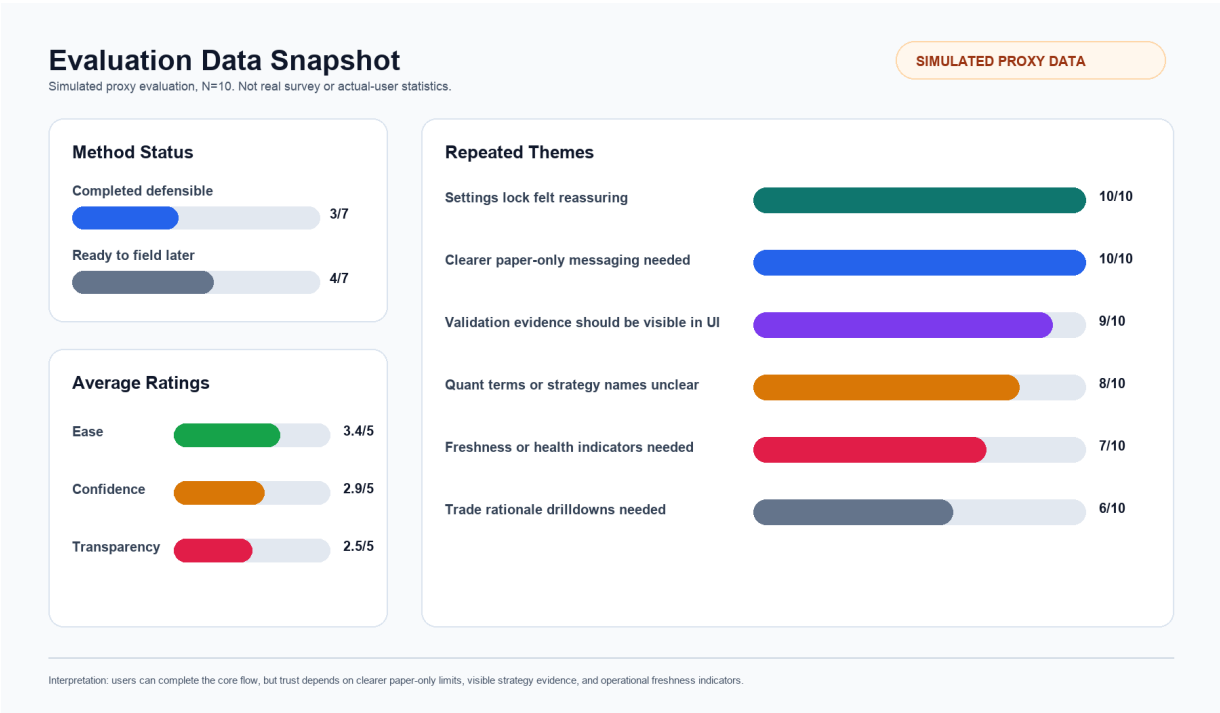


Figure 1. Summary visualization from simulated proxy evaluation data, N=10. Not real survey statistics.

Metric	Value	Meaning
Completed defensible methods	3 / 7	Expert/self evaluation, cognitive walkthrough, simulated user focus group.
Ready-to-field methods	4 / 7	Actual users, low-fi usability, survey, and high-fi usability protocols prepared.
Average ease rating	3.4 / 5	Users can likely complete the core flow.
Average confidence rating	2.9 / 5	Trust is moderate and needs stronger proof.
Average transparency rating	2.5 / 5	Evidence visibility and explanation are the biggest weaknesses.
Paper-only clarity requested	10 / 10	All proxy personas wanted clearer no-live-orders messaging.
Strategy evidence requested	9 / 10	Most proxy personas wanted DSR, PBO, benchmark, and report provenance in the UI.

Table 2. Headline data from the simulated proxy review and evaluation plan.

4. Evaluation Findings By Method

4.1 Expert / Self Evaluation

The expert review found that O(Alpha) is suitable as a controlled paper-trading prototype and research-to-paper workflow demo. It should not be presented as a live-trading product or as self-evidently validated without stronger in-product evidence surfaces.

Feature	Rating	Analysis	Suggested improvement
Product scope	4/5	Research and paper trading scope is clearly safer than live trading.	Repeat paper-only scope near risky-feeling actions.
Onboarding	3.5/5	Risk profile, strategy choice, backtest, and acceptance create a coherent setup path.	Explain what the backtest does and does not prove.
Settings safety	4.5/5	Active-agent settings lock is strong and consistent with fail-closed safety.	Keep backend as source of truth; add user-friendly reason copy.
Dashboard observability	3/5	Dashboard categories are right, but freshness and health state are not prominent enough.	Add latest bar date, last snapshot, heartbeat, and stream state.
Strategy evidence	2.5/5	Validation evidence is mostly in docs/reports instead of user-facing cards.	Add evidence drawer with DSR, PBO, benchmark, OOS trades, costs, and report path.

4.2 Cognitive Walkthrough / Heuristic Evaluation

Task / heuristic	Rating	Analysis	Suggested improvement
Onboarding completion	3.5/5	Users can select a risk profile, choose a matching strategy, run a backtest, and accept it.	Add beginner explanations for risk, Sharpe, Max DD, and historical simulation limits.
Agent launch / terminate	4/5	The main action is understandable and the lifecycle model is clear.	Add paper-only badges beside Launch Agent and portfolio value.
Settings while active	4.5/5	Settings are locked while a portfolio agent is running, preventing confusing mid-run changes.	Keep controls disabled and add concise reason text.
Evidence discovery	2/5	Users must leave the product to verify promotion artifacts and report lineage.	Add report links and validation metadata directly to strategy cards.
Dashboard status	3/5	The dashboard shows portfolio categories but not enough stale/error/last-updated state.	Add freshness, active run ID, heartbeat, and stream status.
Copy clarity	3/5	Some labels are polished but less direct than users need.	Use plain labels such as Save settings, Saving, Settings locked, and Run backtest.

5. Simulated User / Persona Testing

Evidence boundary

The table below represents simulated proxy personas, not actual users. It is useful for planning product improvements and future research, but it should not be labelled as real user testing.

Ratings below are overall averages across ease, confidence, and transparency signals.

User / persona	Feature tested	How they felt	Analysis	Suggestions for improvement
Beginner paper trader	Onboarding, dashboard, launch	3.3/5	Could complete onboarding and launch a paper agent, but was unsure about Sharpe, Max DD, h63, LGBM, and demo/live distinction.	Add tooltips and visible Paper only / no brokerage orders labels.
Risk-aware retail investor	Strategy proof, trades, settings	2.7/5	Liked safety boundaries but could not fully verify strategy quality from the UI.	Add evidence-aware strategy cards and trade audit drawers.
Quant hobbyist	Backtest evidence, catalog proof	3.0/5	Understood the workflow but wanted DSR, PBO, costs, benchmarks, and report links.	Show Promote, DSR, PBO, benchmark, OOS trades, costs, and report path.
Skeptical product manager	Onboarding narrative, proof	3.0/5	Saw a coherent product story but felt proof was mostly in docs rather than screens.	Add a plain-language proof panel after onboarding and on the dashboard.
Reliability-focused engineer	Lifecycle, health, failure state	3.3/5	Trusted the settings lock, but wanted more operational diagnostics.	Add last refresh, active run ID, heartbeat, stream state, and failure reason.
Finance student	Learning, backtest interpretation	3.0/5	Found onboarding approachable but might read Accept Backtest as this strategy is good.	Change acceptance copy to acknowledge historical paper simulation limits.
Compliance-minded reviewer	Claims, disclaimers, audit trail	2.5/5	Wanted claims to match artifact-backed evidence and paper-only limits.	Add compliance-safe no-advice, no-live-order, and artifact lineage labels.
Active trader / power user	Trade activity, exports, drilldowns	2.8/5	Wanted faster inspection of trades, alerts, and allocation changes.	Add trade detail drawer with target weight, previous/current position, fill price, timestamp, and related alert.
Cautious nontechnical saver	Safety, plain language	2.7/5	Liked guardrails but felt the product was technical and could over-trust the backtest.	Add plain-language safety mode and clearer empty/demo states.
Privacy-conscious evaluator	Seeded demo, reproducibility	3.0/5	Wanted to review without exposing personal financial data.	Create deterministic demo script with seeded users and paper-only portfolios.

6. Theme Counts And Product Interpretation

Theme	Proxy count	Product interpretation
Settings lock felt reassuring	10 / 10	Keep the active-agent safety guardrail as a core product proof point.

Theme	Proxy count	Product interpretation
Clearer paper-only messaging needed	10 / 10	Repeat no-live-orders language near every risky-feeling action.
Validation evidence should be visible in UI	9 / 10	Expose strategy proof rather than requiring users to inspect repo reports.
Quant terms or strategy names unclear	8 / 10	Add tooltips and beginner-mode explanations.
Freshness or health indicators needed	7 / 10	Make backend freshness, stream status, latest bar date, and heartbeat visible.
Trade rationale drilldowns needed	6 / 10	Explain why a simulated trade happened and how allocation changed.

7. Recommended Improvements

Priority	Area	Recommendation	Reason
P0	Paper-only clarity	Add persistent Paper only / no brokerage orders labels near Launch Agent, P&L, onboarding backtest acceptance, and settings.	Reduces accidental over-trust and improves evaluator confidence.
P0	Strategy evidence	Add evidence-aware cards or drawers with promotion status, DSR, PBO, benchmark, OOS trades, cost stress, report path, and model artifact requirements.	Brings the project's strongest validation story into the product UI.
P1	Demo and empty states	Replace realistic fallback values with explicit demo, unavailable, or empty states.	Prevents missing backend state from looking like real performance.
P1	Dashboard health	Show latest bar date, last portfolio snapshot, last agent evaluation, active run ID, heartbeat, and stream connection state.	Improves reliability trust and operational transparency.
P1	Trade audit drawer	Show triggering strategy, target weight change, previous/current position, simulated fill price, timestamp, and related alert.	Answers the common why did this happen question.
P2	Education layer	Add tooltip explanations for Sharpe, Max DD, DSR, PBO, active sleeve, regime, LGBM, h63, leverage, stop-loss, and take-profit.	Makes the product more approachable for beginner users.

8. Future Real-Participant Research Plan

Method	Target sample	Privacy-safe setup	Status
Actual user focus group / interviews	10 participants	45-60 minute private sessions with seeded demo accounts; no real portfolio data.	Not completed.
Low-fidelity usability testing	5-8 participants	Figma or wireframe tests for onboarding, evidence drawer, dashboard status, and settings lock.	Not completed.

Method	Target sample	Privacy-safe setup	Status
Survey of potential users	50 respondents	Private survey covering trust, safety expectations, strategy evidence, and feature priorities.	Not fielded.
High-fidelity usability testing	8-12 participants	Working prototype with seeded account; measure task success, assists, trust, safety comprehension, and SUS.	Not completed.

The best next evidence step is a private high-fidelity usability test with five real participants using seeded accounts. This would provide stronger evidence than additional simulated personas while avoiding public launch and protecting privacy.

9. Simulated Proxy Survey Results

Evidence boundary

The results below come from a simulated 50-person proxy survey run through subagents. They are useful as fast product-planning and investor-narrative signals, but they are not real survey responses and should not be represented as actual user research.

The proxy survey was segmented into 20 beginner/intermediate potential users, 15 risk-aware or technical potential users, and 15 evaluator/compliance/finance-student personas. Ratings are weighted by segment size.

Simulated survey metric	Weighted result		Interpretation
Concept clarity	4.1 / 5		The concept is easy to understand when framed as research-first paper trading.
Paper-only understanding	4.4 / 5		The paper-only positioning lands well and should be repeated visually in product.
Trust after safety guardrails	4.1 / 5		Active-agent locks and no-live-trading positioning are strong trust builders.
Desire for validation evidence	4.7 / 5		Report-backed strategy evidence is the strongest conversion driver.
Interest in beginner education	4.2 / 5		Guided explanations remain valuable, especially for beginner and student audiences.
Likelihood to try private demo	3.9 / 5		Private demo interest is positive but depends on clarity, simplicity, and proof.
Proxy segment	N	Segment signal	
Beginner/intermediate investing users	20	Concept clarity 4.1; paper-only 4.4; trust 4.2; validation evidence 4.6; education 4.5; demo likelihood 3.9.	
Risk-aware / technical users	15	Concept clarity 4.1; paper-only 4.5; trust 4.3; validation evidence 4.8; education 3.7; demo likelihood 4.0.	
Evaluator / compliance / finance-student users	15	Concept clarity 4.1; paper-only 4.4; trust 3.9; validation evidence 4.7; education 4.3; demo likelihood 3.8.	

Strongest positive signal: the product is compelling when positioned as a paper-only research lab with visible safety controls and report-backed strategy evidence. Main funding-readiness gap: actual demo usage, real participant feedback, and credible proof that users return after the first session.

- Simulated quote: "The safety-lock framing makes it feel more like a research lab than a trading bot, which is the right direction."
- Simulated quote: "Report-backed validation is the part I care about most. Show me the evidence and I'll spend time with it."
- Simulated quote: "If it stays clearly paper-only and teaches the reasoning behind strategies, I can see this being useful for finance students without encouraging risky behavior."

10. Potential User Survey Instrument

Survey status

A real potential-user survey has not been fielded yet. The instrument below is ready to use with 50 real potential users, but real-world results should only be reported after actual responses are collected and anonymized.

Module	Survey item	Response type	Why it matters
Screenener	How would you describe your investing or trading experience?	Beginner / intermediate / advanced / none	Segments results by experience level.
Screenener	Have you used a paper-trading, investing, or backtesting tool before?	Yes / no	Separates familiar users from first-time users.
Concept clarity	I understand that O(Alpha) is for paper trading only.	1-5 agreement	Checks whether the safety scope is understood.
Trust	I would want to see validation evidence before launching an automated strategy.	1-5 agreement	Tests demand for report-backed proof.
Reverse-coded trust	A five-year backtest alone is enough for me to trust a strategy.	1-5 agreement	Flags over-trust in simple backtests.
Safety concern	I am concerned about accidentally placing real trades.	1-5 agreement	Measures need for stronger no-live-orders messaging.
Evidence	I would find DSR, PBO, benchmark, and report links useful on strategy cards.	1-5 agreement	Validates the evidence-card roadmap.
Education	I would use short explanations for terms like Sharpe, drawdown, PBO, and DSR.	1-5 agreement	Validates beginner education features.
Dashboard value	Rank the most important dashboard information.	Ranking	Prioritizes P&L, positions, trades, alerts, strategy proof, and data freshness.
Open response	What would make you trust this product more?	Open text	Captures unmet trust requirements.
Open response	What would make you stop using this product?	Open text	Captures blockers and safety concerns.

11. Honest Rating Positioning For Submission

The current ratings should remain as recorded. A stronger funding or submission story can still be made by separating current evidence from target outcomes after the recommended improvements. The target column below is a roadmap goal, not a claimed result.

Metric	Current evidence	Target after improvements	How to earn the target
Ease	3.4 / 5	4.0+ / 5	Add tooltips, clearer backtest acceptance copy, and beginner-mode explanations.
Confidence	2.9 / 5	4.0+ / 5	Surface paper-only labels, proof panels, and report-backed strategy evidence.
Transparency	2.5 / 5	4.1+ / 5	Add DSR, PBO, benchmark, OOS trades, costs, report path, and dashboard freshness indicators.
Simulated survey appeal	4.1-4.7 / 5	Validate with real users	Field the prepared 50-person survey and compare real responses to the proxy benchmark.
Safety clarity	10 / 10 proxy theme	Maintain 10 / 10	Keep active-agent lock, backend fail-closed guard, and explicit no-live-order messaging.

Appendix: Source Artifacts

- docs/evaluation/solution_suitability_evaluation_2026-07-05.md
- docs/evaluation/solution_suitability_visual_summary_2026-07-05.md
- docs/evaluation/solution_suitability_metrics_2026-07-05.csv
- docs/evaluation/solution_suitability_charts_2026-07-05.svg
- docs/evaluation/simulated_proxy_survey_results_2026-07-08.csv
- docs/evaluation/potential_user_survey_questionnaire_2026-07-08.csv
- README.md, docs/submission/OALPHA_README.md, onboarding, dashboard, settings, activity, and strategy catalog evidence files.